Data Structures & Algorithms (CST-003) - Unit 5

Unit 5 - Sorting & Hashing - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: the standard sorting algorithms and hashing - how to order data and how to find it in near-constant time.

Key ideas: Simple sorts: **bubble** (swap neighbours, $O(n^2)$), **selection** (pick the minimum each pass, $O(n^2)$), **insertion** (grow a sorted prefix; fast on nearly-sorted data). Fast sorts: **merge sort** (divide, sort halves, merge - always O(n log n), needs extra space) and **quick sort** (partition around a pivot - O(n log n) average, $O(n^2)$ worst on sorted input with bad pivots). **Hashing** maps a key to an index with a hash function (division method: key % tableSize). Collisions are handled by **chaining** (each slot holds a list) or **open addressing** (linear probing: try next slot).

Exam pointers: one full dry run of a sort (usually quick or merge) showing each pass is the repeated numerical; a complexity table of all five sorts (best/average/worst) earns quick marks. For hashing, practise building a table for given keys with linear probing and with chaining.

Made by Anas. Free forever. Watch the linked videos on the site after reading.